

"Combustion"

Simplification

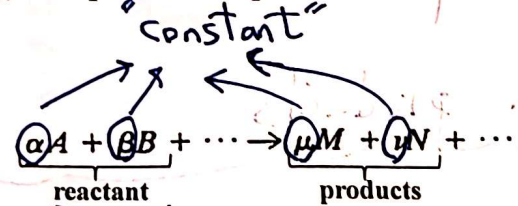
- Chemical reactions nearly always occur between gases that can be approximated a perfect gas. chemical
- Except very rapid expansion processes, the chemical processes usually take place approximately in equilibrium.

❖ Determine the composition and temperature of the exhaust products of a constant-pressure adiabatic combustion process.

To formulate methods :

- (a) chemical transformation and (b) composition of products.

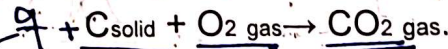
A. Chemical Transformation



chemical equ. means that there is no change in concentration of reactants & products.

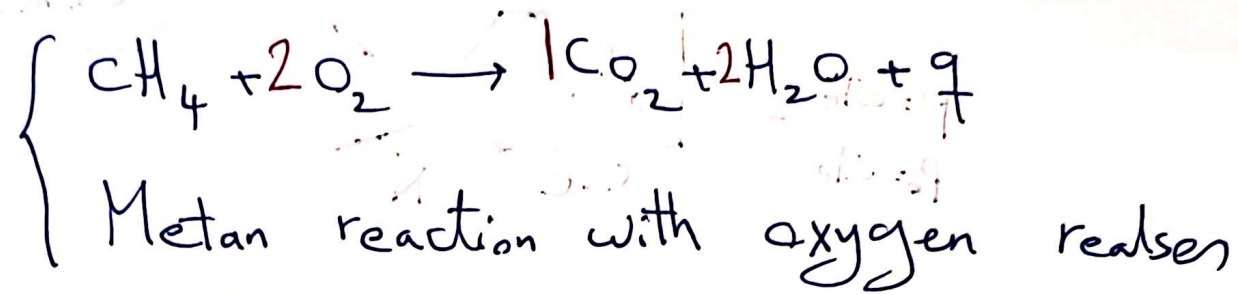
The constant- pressure heat of formation of a substance is the heat interaction that occurs when one mole of the substance is formed at constant pressure and temperature from its elements as they occur in nature.

- Heat of formation of carbon dioxide (-393.522 kJ/gmol) at 298 K:



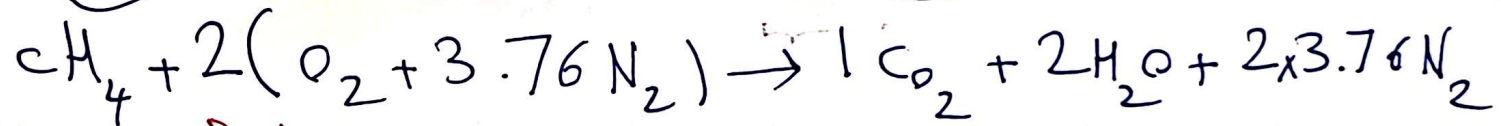
- System must give off heat to maintain (or return to) the reference temperature.

← endothermic
"absorbs heat"



heat, it is exothermic

Stoichiometric ratio: it refers to the exact air-to-fuel ratio needed for complete combustion. (No excess air, no excess fuel) in products



Air to fuel ratio

$$(AF)_{sto} = \frac{\text{Mass of air}}{\text{Mass of fuel}} = \frac{2 \times [32] + 3.76 \times 2 \times 28}{12 + 4} = 17.16$$

Air

molar mass of N₂

$AF > AF_{sto} \rightarrow$ Air-rich reaction
 $AF < AF_{sto} \rightarrow$ fuel-rich reaction

Stoichiometric air to fuel ratio

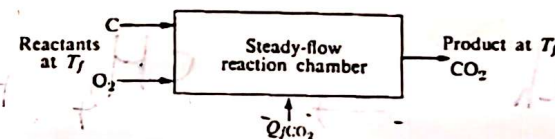
❖ Determine heat quantities:

- Experimentally with a simple constant-pressure container: difficult to produce a reaction at constant temperature and pressure.
- A steady-flow process (experimental): heat transfer is equal to the change in enthalpy. (No shaft work)

1st law
 $W=0$
 $Q = \Delta H$

$$Q = Q_f = H_{\text{product}} - H_{\text{reactants}}$$

$Q_f < 0$ endothermic
 $Q_f > 0$ exothermic



- The heat of formation of naturally occurring elements : zero.
- Complex reactions: sum of distinct decomposition and formation reactions.

$$Q_1 = - \sum (n_i Q_{fi})_{\text{reactants}}$$

number of moles of reactant i

heat of formation per mole

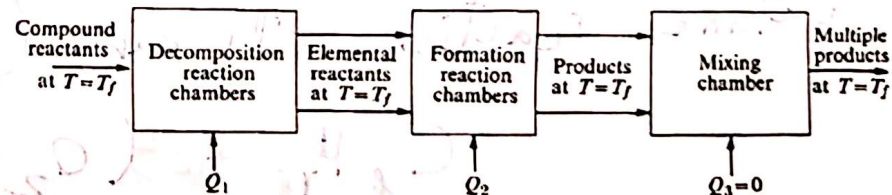
❖ Once the reactants are in elemental form, we may treat the remainder of the reaction as the sum of separate formations, so that

$$Q_2 = \sum (n_j Q_{fj})_{\text{products}}$$

$$Q_R = Q_1 + Q_2 = \sum (n_j Q_{fj})_{\text{products}} - \sum (n_i Q_{fi})_{\text{reactants}}$$

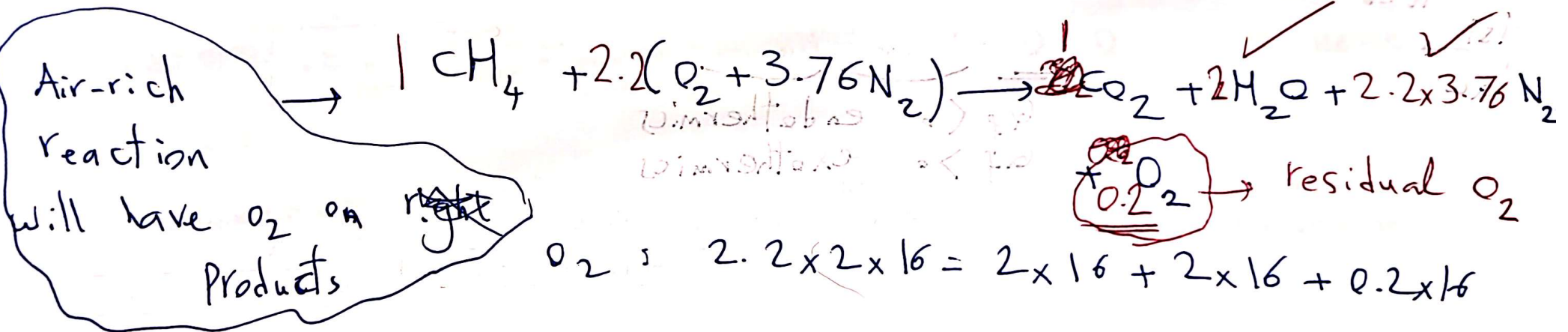
$$Q_R = H_{\text{products}} - H_{\text{reactants}} = \sum (n_j Q_{fj})_{\text{products}} - \sum (n_i Q_{fi})_{\text{reactants}}$$

Net heat of interaction at constant temperature

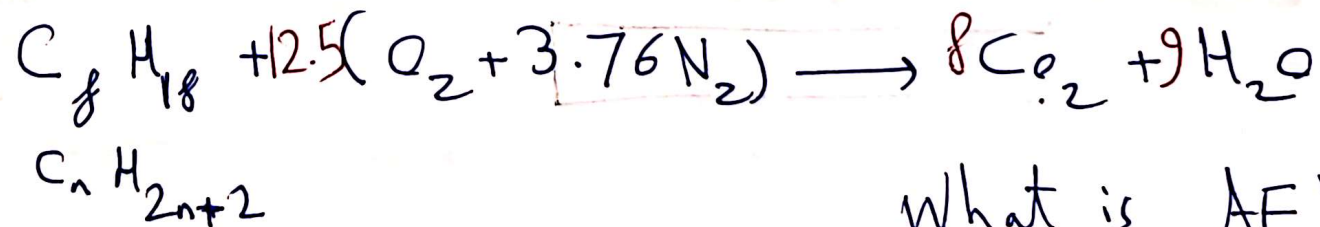


if you have 10% air-rich reaction, then compute n_{air}
(Number of moles of air)

$$AF = 1.1 \times 17.16 = \frac{n_{air} [32 + 28 \times 3.76]}{16} \Rightarrow n_{air} = 2.2$$



octane reaction with air (stoichiometric)



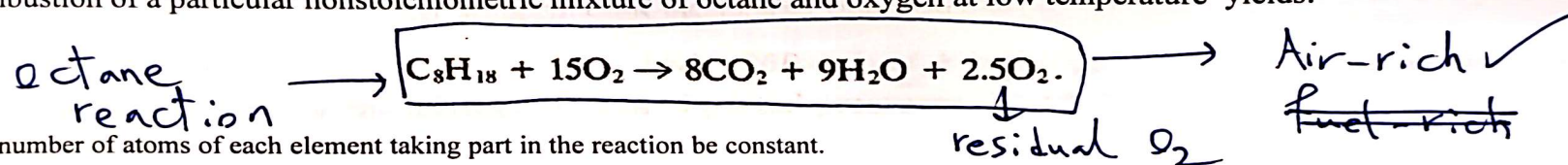
What is $AF)_{sto}$?

(Always start with stoichiometric) $\Rightarrow$

❖ Composition of Products:

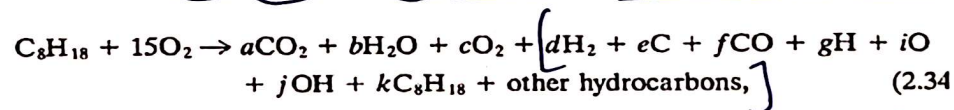
The product composition depends on the temperature, which in turn has just been shown to depend on the composition. This suggests that we may need to use a trial-and-error procedure for the solution.

❖ combustion of a particular nonstoichiometric mixture of octane and oxygen at low temperature yields:



✓ the number of atoms of each element taking part in the reaction be constant.

❖ At high temperatures the compounds CO_2 , H_2O , and O_2 dissociate to some extent into free atoms and radicals:



✓ all practical purposes C_8H_{18} and the "other hydrocarbon" molecules are present in negligible

❖ Atom balances yield:

For <u>carbon</u> ,	$a + e + f = 8;$
For <u>hydrogen</u> ,	$2b + 2d + g + j = 18;$
For <u>oxygen</u> ,	$2a + b + 2c + f + i + j = 30.$

six more independent equations require.

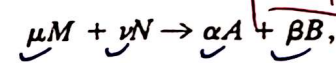
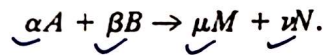
$$AF)_{sto} = \frac{12.5 (32 + 3.76 \times 28)}{8 \times 12 + 18} = \frac{1739.13 \text{ g}}{114 \text{ g}} = \boxed{15.25}$$

diagram ←  ← notes
 constant 2 indices

$$\frac{[\text{partial pressure of product M}]^{\mu} \times [P_N^v]}{[\text{reactant A}]^{\alpha} \times [P_B]^{\beta}} = K_p(T)$$

❖ Chemical equilibrium:

Stoichiometric combination of A and B react to form products M and N:

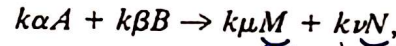


Equ. constant of reaction

➤ A chamber that communicates through semipermeable membranes with each individual consistency substance in a pure form.

- Stoichiometric ratio:

❖ Equilibrium constant for the mixture in term of partial pressures.



small number

a heat interaction of magnitude kH_{RP}

$$\frac{p_M^{\mu} p_N^{\nu}}{p_A^{\alpha} p_B^{\beta}} = K_p(T)$$

in chemical Equ., the following ratio is constant

$$K_n = \frac{\chi_M^{\mu} \chi_N^{\nu}}{\chi_A^{\alpha} \chi_B^{\beta}} = \frac{\chi_M^{\mu} \times \chi_N^{\nu}}{\chi_A^{\alpha} \times \chi_B^{\beta}}$$

❖ Based on mole fractions :

According to the Gibbs-Dalton law,

$$\chi_i = p_i/p_m$$

$$K_n = p_m^{\alpha+\beta-\mu-\nu} K_p$$

relation between K_n & K_p

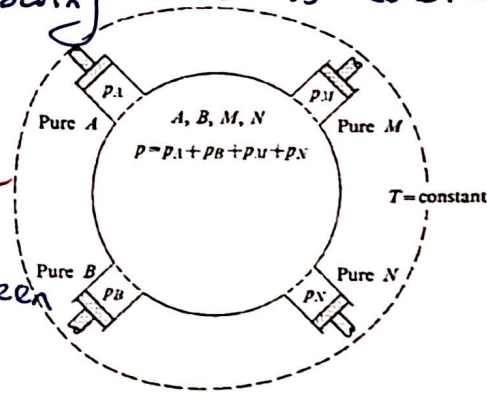
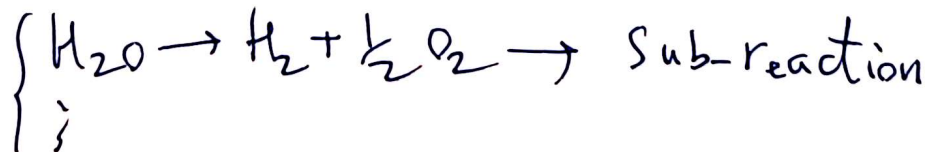


FIGURE 2.11 Reaction chamber for derivation of equilibrium constants.

➤ One of the six relationships necessary to the determination of product composition.

➤ A large Equilibrium constant for subreactions that approach completion and a small for those that do not.



$K_p \uparrow \rightarrow \text{occurs}$
 $K_p \downarrow \rightarrow \text{Not occurs}$

❖ Reactions Close to Stoichiometry (Rocket combustors , Ramjet combustors, Piston engines, Welding torches)

- Generate very high gas temperatures
- Materials limits are either absent or can be overcome by appropriate cooling of walls
- Cannot easily predict the final composition of the burnt gases
- complex molecules like H_2O tend to break down into simpler ones such as O, H, OH

➤ Determine the temperature and other properties of the gas products:

- I. Mass conservation
- II. Energy conservation
- III. Laws of chemical equilibrium

❑ Iteration procedure:

1. Identify the possible molecular species in the products, classify them into major and minor.
2. Neglect at first the "minor" species, use procedures similar to those in the previous example to obtain the "major" species concentration and an approximate temperature.
3. Write chemical reactions which would produce one "minor" species at a time from only "major" ones. Calculate their equilibrium constants at the approximate temperature of step (2), and use the equilibrium laws to calculate the concentrations of the "minors".
4. Correct the "majors" concentration, and do a new enthalpy balance to obtain an improved temperature.
5. Iterate back to step (3) until convergence is obtained